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भारतीय मानक मसौदा
अल्प शल्य चिकित्सा के लिए द्रव-चालित मेज - विशिष्टि
[IS 6106 का पहला पुनरीक्षण]

Draft Indian Standard
Minor Operation Hydraulic Table - Specification
[First Revision of IS 6106]

ICS 11.140

Hospital Equipment and Surgical Disposable
Products Sectional Committee, MHD 12

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FOREWORD

(Formal clause will be added later)

This standard was originally published in 1971. The first revision of this standard has been brought out to align it with recent developments in the material requirements, addition of tolerance to the dimensions, removal of wooden handle and to bring the standard in line with the latest style and format of referred standards.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2: 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This standard specifies the construction and performance requirements for manually controlled height minor operation hydraulic tables.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

IS No.	Title
IS 6911: 2026	Stainless Steel Plate, Sheet and Strip - Specification (<i>third revision</i>)
IS 1068: 2025 ISO 1456: 2009	Metallic and other inorganic coatings - Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and of copper plus nickel plus chromium (<i>fourth revision</i>)
IS 17523: 2021	Castors and Wheels - Requirements for Castors for Furniture

3 TERMINOLOGY

3.1 Minor Operation Table - A surgical operation table fitted with a hydraulic pump or mechanical device (or electro-mechanical device) to raise and lower.

3.2 Jerk - The sudden change in speed or quick movement.

4 MATERIAL REQUIREMENTS

4.1 The materials used in the manufacture of the minor operation table shall be corrosion-resistant or protective finished.

4.2 The raising column and the gearing system shall be made from materials to resist wear during normal manipulation.

4.3 The handles provided shall be made from ebonite or phenol formaldehyde.

4.4 The stainless steel used shall conform to designation X07Cr17Ni12Mo2 in accordance with IS 6911.

4.5 The minor operation table shall have an antistatic rubber tread or stainless steel castor as per IS 17523. The operation table shall have a static charge transmission mechanism to the earth.

5 SHAPE AND DIMENSIONS

5.1 Table Top

The overall length of the table top shall be 180 ± 5 cm with the head section and 195 ± 5 cm without the head section. The overall width of the table top excluding side railings shall be 50 to 55 cm.

5.2 Height

The height of the table top without mattress shall be 80 ± 5 cm and in the extreme lifted position preferably between 95 ± 5 cm subject to limitation of the hydraulic lift as specified in **6.4**.

6 CONSTRUCTION REQUIREMENTS

6.1 Design

A typical design for the minor operation table shall be as per Fig. 1.

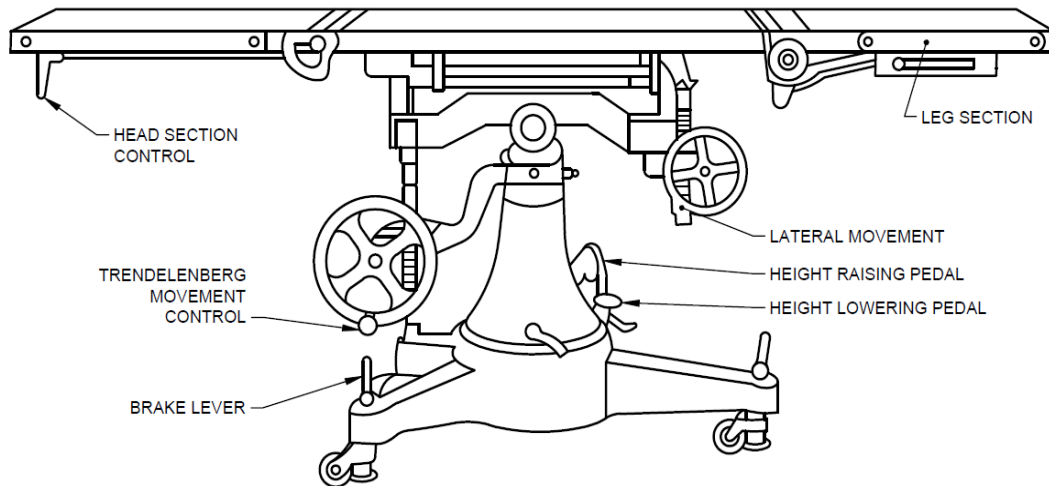


Fig. 1 MINOR OPERATION HYDRAULIC TABLES

6.2 Base

The base shall be solid and mounted on castors to provide stability to the minor operation table. The base shall have non-skid compensating floor locks actuated by a foot pedal. The base may also have foot rails for providing a comfortable footrest for the surgical team.

6.3 Table Top

The table top shall have a minimum of three sections, namely, head, trunk and leg sections. The head section shall be detachable. The detachable feature for the leg section depends on manufacturer and purchaser requirements. The entire table top shall have a jerk-free vertical movement.

The multiple cylinder hydraulic systems may have sudden change in speed (jerk) as a result of cylinder diameter differences which shall not be in excess of ' $n - 1$ ', where ' n ' is the number of cylinders. The minor operation table shall also satisfy the following requirements:

- The entire table top shall be capable of being tilted to $30^\circ \pm 1^\circ$ in Trendelenburg and reverse Trendelenburg positions.
- The table shall have arrangements for a lateral tilt of $20^\circ \pm 2^\circ$ in either direction.
- The table top may be provided with a perineal cut-out for the drainage tray.

6.3.1 Head and Leg Sections

The head section shall be able to raise 40° to 45° and in its reverse position 90° from the trunk portion. The leg section shall be capable of being raised or lowered to 90° from the trunk to enable the table to be used as a chair. The head section and the table top shall have controls capable of positioning during operation and the controls shall be situated so as not to interfere with the surgical team. The various controls may be independent and on sides of the table with removable type or folding type handles.

6.4 Hydraulic Lift

The table shall have a hydraulic pump or mechanical device (or electro-mechanical device) lifting/lowering device fitted with the base of the table. The table shall operate by a paddle or a handle. The table lift shall be between 15 to 25 cm. The controls shall have easy access and various component parts of the controls and the table shall be easy to clean and maintain. All parts intended to be removed by the user shall be easy to replace with parts of the same make and difficult to assemble incorrectly.

7 ACCESSORIES

7.1 The Table 1 shall be provided with the following standard accessories:

Table 1 List of Accessories

Sr. No.	Accessories
1	Foot extension
2	Shoulder supports, Side supports with antistatic rubber pads
3	Anaesthetic screen
4	Leather Wristlets
5	Pair of knee crutches/lithotomy leg holders
6	25 to 50 mm thick antistatic foam rubber mattress

NOTE - When antistatic rubber is not used, the construction of the table shall ensure transmission of static charge to the earth.

8 PERFORMANCE REQUIREMENTS

8.1 Hydraulic Lift

The hydraulic lift shall give a vertical jerk-free and smooth movement to the system. When the hydraulic pressure is released, the table top shall take not less than 10 seconds and not more than 30 seconds to travel from the maximum height to the bottom most position. The downward movement shall be smooth and subject to limitations in 6.3. When the table top is held in any position by hydraulic action alone, and loaded with 150 kg in the trunk portion, it shall not sink by more than 3 mm in a period of four hours.

8.2 Base

The base shall be stable on the castors with non-skid floor locks. The table shall be applied with 120N load to move the table on a smooth cemented floor after the table is moved 70N the load shall be sufficient for the motion. The floor locks shall give rigidity to the table and the table shall not move in any direction when a force of 500N is applied to the base in any direction.

8.3 Table Top

The table top shall satisfy the requirements given below in any position obtained by the hydraulic lift:

- The extreme edge of the head section shall be applied with a vertical load of 500N, when the load is applied the leg section shall not lift by more than 15 mm.
- The extreme end of the head section (with the head section drawn in) shall be applied with a load of 5 kg which is dropped from a height of 150 mm, when the load is applied the head section shall not get detached.
- The head section shall be pulled along its plane with a load of 250N, when the head section is locked it shall not come off.
- There shall be no wobble under normal working conditions when the accessories are clamped.

9 WORKMANSHIP AND FINISH

9.1 The minor operation table components shall provide smooth operation and additional performance requirements as per **8**. The operation table surfaces shall be well finished and free from surface defects, such as pinholes, blisters, blow-holes and similar other defects. The edges shall be rounded off and the three sections comprising the table top in their normal position shall be in one plane when tested with a spirit level. The sweeps of the head and leg sections shall be easy. The welded joints shall have welds fully penetrating and well finished. The minor operation table metal surfaces shall be non-corrosive (aluminium and stainless steel) with matt finish.

9.2 The corrosive surface like steel castings for base, cylinder casings, foot pedals shall be painted to shades such as sea green or low reflective shades agreed to between the manufacturer and the purchaser. Initially the surface which is painted shall be degreased, rust-proofed by phosphating and then suitably protected by an anti-corrosive primer with brushing or spraying and then finished by spraying in stove enamel or air-drying enamel of the specified shade. During the procedure each stage shall be separately stoved or air-dried.

9.3 The resulting finish shall be hard and shall not readily chip or flake. The mild steel and brass parts shall be plated chromium over nickel according to **Service Condition 2** of IS 1068.

10 MARKING

10.1 The marking which is used by the manufacturer shall be legibly and indelibly marked with the Identification mark of the manufacturer.

10.2 Each minor operation table shall have a name plate fixed to it showing the following:

- a) Range of hydraulic lift;
- b) Any special instructions for safe handling of the table.

10.3 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

11 OPERATING MANUAL

11.1 The minor operation table shall be accompanied by instructions manual giving the following information:

- a) Brief instructions for installation and maintenance (including spare parts), and
- b) Instructions for correct operation of the table.

12 PACKAGING

The minor operation table shall be packed as agreed to between the purchaser and the manufacturer. Each individual package shall contain only one unit and may be marked with the name of the item, and identification of the source of manufacture and batch number and any other marking if so desired.